

Predicting and Explaining Employee Absenteeism at Work

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I. Original Project Idea and Plan

The goal of this project is to analyze the factors that influence employee absenteeism at work and to build predictive models that can predict absenteeism time in hours. Employee absenteeism is an important workforce issue because frequent or prolonged absences can disrupt operations, reduce productivity, and increase labor costs. Using a real-world absenteeism dataset¹ from a courier company in Brazil (Martiniano et al., 2012), this project will examine and possibly explain if there is a relationship between demographic, lifestyle, transportation, work-related, and medical factors associated with absence duration.

The dataset we plan to use is the Absenteeism at Work dataset¹, which contains records of absenteeism from July 2007 to July 2010 at a courier company in Brazil, downloaded from the UC Irvine Machine Learning Repository. Since this is an existing public dataset, no new data collection is required; instead, the data will be downloaded directly from the UCI repository and imported into Python for cleaning, exploration, and modeling.

The dataset contains 740 records and 21 attributes, with **Absenteeism time in hours as the target variable**. The features cover absence reasons, time-related factors, transportation, work history, age, workload, education, family information, health-related measures, and lifestyle habits such as drinking and smoking².

II. Project Updates and Modifications

After revisiting the original neuro-fuzzy paper and our dataset more carefully, we decided not to make direct model comparison the main direction of the project. Although the paper is the source of the dataset and motivates our topic, it is not fully replication-ready for a fair benchmark. The paper describes the general neuro-fuzzy architecture and states that mean absolute error was used as the evaluation metric, but it does not provide a clear numeric MAE result, detailed reproducible preprocessing steps, or a sufficiently specific experimental setup for direct comparison (Martiniano et al., 2012). As a result, a strict model-to-model comparison would be difficult to justify.

¹ *Absenteeism at Work*, UCI Machine Learning Repository, <https://archive.ics.uci.edu/dataset/445/absenteeism+at+work>

² Due to page limitations, we do not list all 21 attributes (and their descriptions) here.

Based on this limitation, we refined our project direction toward interpretable predictive modeling of absenteeism time in hours. In addition to the original neuro-fuzzy study, later work on absenteeism prediction has applied data mining and tree-based learning methods, which helped motivate our choice of predictive modeling approaches in this project (Skorikov et al.; Wahid et al.). Instead of focusing on reproducing the original neuro-fuzzy model, we now emphasize exploratory analysis and modern regression approaches that can better explain which factors are associated with longer absences. In particular, we compare an interpretable linear baseline (ridge regression) with a more flexible nonlinear model (XGboost regression), while examining the effects of medical absence reasons, lifestyle indicators, and work-related variables on absenteeism. This updated direction remains consistent with our original proposal, which emphasized interpretable analysis and identifying the factors linked to longer absence duration.

Additionally, although clustering was briefly mentioned in the work division section of our proposal, it was not clearly introduced in the main project description. After examining the dataset more carefully, we found that it would also be interesting to analyze absenteeism at the employee level rather than only at the individual absence-record level. Since the dataset contains repeated records for a small set of employees, we can construct employee-level profiles by aggregating information for each ID, such as average absence hours, fraction of longer absences, distribution of absence reasons, and average health- and work-related characteristics like BMI and workload. Based on these profiles, we may further explore clustering employees into different absenteeism patterns using methods such as k-means or hierarchical clustering, while checking cluster stability carefully given the small number of employees.

III. Progress and Initial Results

We have completed data preprocessing, exploratory analysis, and supervised learning experiments. The following summarizes the key findings.

During exploratory analysis, we found that absenteeism hours are strongly right-skewed. Based on the IQR rule, we identified 17 hours as the upper bound for outlier detection and therefore examined the reasons associated with absences greater than or equal to 17 hours.

Among these longer absences, the most common categories were Reason XIII — Diseases of the musculoskeletal system and connective tissue (29.55%) and Reason XIX — Injury, poisoning and certain other consequences of external causes (22.73%). These reasons are medically plausible for longer absence durations because they often involve conditions that require longer recovery periods.

In addition to the distribution of the absenteeism hours, we identified that the dataset contains absence records for only 36 employees across 740 observations. Because the same employee appears multiple times, a standard random train-test split could allow information from the same individual to appear in both training and testing sets, leading to overly optimistic performance estimates. To address this issue, we adopt group-aware validation based on employee ID.

The initial regression experiments using Ridge Regression and XGBoost Regression showed relatively large prediction errors when modeling all absenteeism durations together, with average 5-fold GroupKFold MAE values of 7.89 and 5.69 hours, respectively. This suggests that the target variable is difficult to predict across its full range and that extreme high-hour cases may behave differently from more typical absences.

These early results indicate that the two models may not fully capture the structure of the data, so the next stage of the project will examine error patterns more closely and test whether performance improves after sensitivity analysis with exclusion or separate treatment of long absences as mentioned above. This direction is also supported by more recent work on the same dataset, which notes the strongly right-skewed nature of absenteeism duration and excludes cases above 30 hours when focusing on typical absence patterns (Llamas Blázquez, 2025).

We also found that absence durations greater than 8 hours are typically recorded in multiples of 8, suggesting that longer absences may follow a discrete reporting pattern such as full-day or multi-day leave. Therefore, we plan to test whether adjusting predictions above 8 hours by rounding them to the nearest multiple of 8 can reduce prediction error and improve alignment between predicted and observed values.

IV. Works Cited

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